

EDUCATION

Udacity, School of Autonomous Systems

Robotics Software Engineer Nanodegree Program

Online

Apr 2021 – Apr 2022

- **Relevant Areas:** ROS, Gazebo, Localization, Mapping, SLAM, Navigation, and Path Planning

Stony Brook University

Ph.D. and M.S. in Mechanical Engineering, GPA 3.95

Stony Brook, NY

Aug 2015 – Aug 2020

- **Concentration:** Design and Robotics, **Minor:** Applied Mathematics
- **Relevant Courses:** Robotics, Advanced Dynamics, Vibration and Control, Kinematic Analysis and Synthesis, Applied Stress Analysis, Product Design Optimization, Geometric Modeling, Analysis of Algorithms

Udacity, School of Autonomous Systems

Self Driving Car Engineer Nanodegree

Online

Mar 2019 – Mar 2020

- **Relevant Areas:** Computer Vision, Deep Learning, Sensor Fusion, Localization, Planning, Control, System Integration

Indian Institute of Information Technology

B.Tech., Mechanical Engineering

Jabalpur, India

Aug 2009 – Aug 2013

- **Relevant Areas:** Mechanical Design and Manufacturing, Computer Aided Design

EXPERIENCE

NVIDIA

Senior Software Engineer - Autonomous Vehicles

Santa Clara, CA

Apr 2026 – Present

- Develop world-model components for autonomous driving by fusing perception and map signals into enriched BEV representations. Enable real-time 3D scene understanding and HD mapless driving across complex urban environments.
- Improved performance of the wait-condition fusion module by 25% through profiling and optimization, reducing disengagements in L2++ driving scenarios.
- Owned Product Lifecycle (PLC) activities across all modules within World Model Context fusion. Delivered two L2++ releases for the Mercedes-Benz program while managing a team of 6 contractors. Ensured compliance with software requirements, SWFMEA, Coverity, code coverage, exception handling, and documentation requirements.
- Developed a scalable geofencing interface for the Lane Event Fusion module across L2++, HFE, L3, and L4 autonomy levels. Integrated map, Most Probable Path (MPP), and Operational Design Domain (ODD) information from upstream modules and provided fused outputs to ODD arbitration, routing, and planning modules.

Lucid Motors

Senior Software Engineer - Maps and Navigation, ADAS/AD

Newark, CA

Nov 2023 – Apr 2026

- Led development of HD map software on NVIDIA Xavier/Orin platforms running QNX, integrating GNSS sensor data with TomTom APIs.
- Delivered ADAS features (Lane Change Assist, Adaptive Curve Speed Control, Hands-Free Highway Assist) through cross-functional collaboration with platform, perception, motion planning, cloud, and field testing teams.
- Enhanced system stability and runtime reliability through memory, CPU, and bandwidth profiling, improving offline performance and reducing error rates.
- Used RoboSense Ruby LiDAR and Oxts GNSS to run LIO-SAM SLAM on short driving clips. Applied Removert for dynamic-object detection and point-cloud filtering to produce static maps. Improved the accuracy and reliability of the autolabeling pipeline.
- Supported software validation for the Lucid Air sedan and Gravity SUV using hardware-in-the-loop and prototype vehicle testing.
- Created interactive visualization and debugging tools with Folium, Plotly, and QGIS, enhancing map data analysis and accelerating issue resolution for ADAS development.

Dematic (Kion Mobile Automation)

Machine Learning Engineer - Perception, Localization and Mapping

Holland, MI

Sept 2020 – Oct 2023

- Developed onboard software for autonomous vehicles (AGVs) in warehouse environments.
- Improved the accuracy and robustness of the reflector extraction algorithm, enabling sub-centimeter localization accuracy.
- Developed a Gazebo-based virtual testing pipeline to improve the line-extraction algorithm using hyperparameter tuning.
- Improved the speed of the feature-based association algorithm by 50%, resulting in a 30% faster localization pipeline. Analyzed real-time CPU utilization of SLAM processes and threads using Valgrind, LTTng, and perf.
- Standardized the pallet pick/drop testing at physical warehouses. Performed accuracy and repeatability analysis using an external FARO laser-tracking system. Also, a procedure to calibrate the lidar, steering encoder, and traction encoder was created.
- Developed a Python-based tool to visualize recorded mapping logs, and automated their offline performance analysis.
- Tackled technical debt in the mapping codebase by reducing complexity, simplifying API, and decoupling tightly connected units. Improved mapping API call latency by creating caching threads.
- Created a commissioning workflow to first map a small warehouse and then localize using the generated map.
- Implemented and integrated human/forklift obstacle detection and pallet detection algorithms using 3D camera data.
- Certified **SAFE (Scaled Agile Framework) Practitioner** and trained to use Scrum, Kanban, and XP in an Agile environment.

Stony Brook University

Research Assistant

Stony Brook, NY

May 2017 – Aug 2020

- Developed a computational framework for data-driven mechanism design innovation supported by a \$450K [NSF grant](#).
- Designed machine learning and algebraic algorithms for the simulation and synthesis of planar, spherical, and spatial single-degree-of-freedom robotic systems, resulting in multiple publications in ASME journals.
- Created [MotionGen](#), a web-based mechanism design framework using the MEAN (MongoDB, Express.js, Angular.js, Node.js) stack and MVC architecture to provide a RESTful web service; developed iOS and Android apps using Apache Cordova.
- Managed the Computer-Aided Design and Innovation Lab and collaborated with a research group of 10+ graduate students.

Teaching Assistant

Aug 2016 – Apr 2017

- Developed [SnappyXO](#), a laser-cut design-driven robotics platform that enables designing mechanisms, structures, and robots. The project raised \$16K+ through an [Indiegogo](#) crowdfunding campaign.
- Advised 250+ students in MEC101 (Freshman Design Innovation), MEC 102 (Engineering Computing), and Vertically Integrated Projects (VIP) Program. The Robot Design projects gained recognition from the Office of President at university.

Vivonics, Inc.

Stony Brook, NY

Product Design Intern

Feb 2016 – Aug 2016

- Coordinated with the design team on developing PMT Monitor, a portable medical headset that detects head trauma.
- Generated concepts for an interpupillary distance adjusting mechanism focusing on manufacturability and robustness.

Leviathan Energy

Stony Brook, NY

Strategic Partnership for Industrial Resurgence (SPIR) Intern

Feb 2016 – May 2016

- Designed and manufactured hydrokinetic turbines with an improved airfoil design that produced 50% more power in collaboration with Leviathan Energy.
- Created engineering models using Solidworks and Autodesk Inventor and fabricated parts using FDM-based 3D printing.

Indian Institute of Information Technology

Jabalpur, India

Junior Research Fellow

May 2014 – May 2015

- Led a \$70K+ research project funded by the Science and Engineering Research Board (SERB), titled “Development of Additive-Subtractive Integrated Rapid Prototyping System for Improved Part Quality.”
- Spearheaded design and manufacturing teams to create a new hybrid 3D printing process using pellet-based screw extruder with CNC machines. Developed toolpath-planning strategies to manufacture CAD models using Hybrid Manufacturing techniques.

Sara Sae Private Limited

Dehradun, India

Design Intern

May 2013 – Aug 2013

- Involved with product development of Hydraulic Tongs used in the oil and gas industry.
- Carried out kinematic and dynamic analysis to find the most probable failure regions.

Altair

Jabalpur, India

Engineering Intern

Sept 2012 – Dec 2012

- Solved practical engineering problems using topology, topography, size, and shape optimization in OptiStruct.
- Achieved 44th rank in Altair All India Optimization Contest 2012.

Vehicle Factory Jabalpur

Jabalpur, India

Manufacturing Intern

July 2011

- Worked with the Production Line and Maintenance Teams for Mine Protected Vehicle and Stallion Mark-I trucks.

Central Institute of Hand Tools

Jalandhar, India

Computer-Aided Design Intern

June 2010

- Performed computer-aided design using Pro/ENGINEER, mold modeling, and mold-flow analysis.

PUBLICATIONS

Ph.D. Dissertation

- Analytical and Machine Learning Based Frameworks for Synthesis of Planar, Spherical and Spatial Mechanisms

Journal Articles

- Sharma S., Purwar A.; **A Machine Learning Approach to Solve the Alt–Burmester Problem for Synthesis of Defect-Free Spatial Mechanisms**, ASME Journal of Computing and Information Science in Engineering 2022; doi:10.1115/1.4051913
- Sharma S., Purwar A.; **Using a Point-Line-Plane Representation for Unified Simulation of Planar and Spherical Mechanisms**, ASME Journal of Computing and Information Science in Engineering 2020; doi:10.1115/1.4046817
- Sharma S., Purwar A., Ge Q.J.; **A Motion Synthesis Approach to Solving Alt-Burmester Problem by Exploiting Fourier Descriptor Relationship Between Path and Orientation.**, ASME Journal of Mechanisms and Robotics 2019; doi:10.1115/1.4042054
- Sharma S., Purwar A., Ge Q.J.; **An Optimal Parametrization Scheme for Path Generation Using Fourier Descriptors for Four-Bar Mechanism Synthesis.**, ASME Journal of Computing and Information Science in Engineering 2019; doi:10.1115/1.4041566

Conference Papers

- Sharma S., Purwar A.; **Path Synthesis of Defect-Free Spatial 5-SS Mechanisms Using Machine Learning.**, ASME 2020 International Design Engineering Technical Conferences and Computers and Information in Engineering Conference; doi:10.1115/DETC2020-22731
- Sharma S., Purwar A.; **Unified Motion Synthesis of Spatial Seven-Bar Platform Mechanisms and Planar-Four Bar Mechanisms.**, ASME 2020 International Design Engineering Technical Conferences and Computers and Information in Engineering

Conference; doi:10.1115/DETC2020-22718

- Sharma S., Purwar A.; **Using a Point-Line-Plane Representation for Unified Simulation of Planar and Spherical Mechanisms**, ASME 2019 International Design Engineering Technical Conferences and Computers and Information in Engineering Conference; doi:10.1115/DETC2019-98194
- Sharma S., Purwar A., Ge Q.J.; **A Motion Synthesis Approach to Solving Alt-Burmester Problems by Exploiting Fourier Descriptor Relationship Between Path and Orientation Data**, ASME 2018 International Design Engineering Technical Conferences and Computers and Information in Engineering Conference; doi:10.1115/DETC2018-85567
- Sharma S., Purwar A., Ge Q.J.; **Optimal Non-Uniform Parametrization for Fourier Descriptor Based Path Synthesis of Four Bar Mechanism**, ASME 2018 International Design Engineering Technical Conferences and Computers and Information in Engineering Conference; doi:10.1115/DETC2018-85568

AWARDS

- Invited by the American Society of Mechanical Engineers (ASME) multiple times to review journal manuscripts.
- Received the Resource Access Project Travel Grant Fall 2019 by Graduate Student Organization, Stony Brook to present research findings at IDETC 2019 in Anaheim, California.
- Received the Resource Access Project Travel Grant Fall 2018 by Graduate Student Organization, Stony Brook to present research findings at IDETC 2018 in Quebec City, Canada.

TECHNICAL PROFICIENCY

- **Robotics hardware** : NVIDIA Jetson (Nano and Xavier NX), 2D and 3D LiDAR (Sick, Ouster, and Velodyne), RGB-D camera (Intel Realsense D455), steering and traction encoder, IMU, Raspberry Pi, Arduino
- **Robotics software** : Keras, TensorFlow, PyTorch, ROS, Gazebo, RViz, Anaconda, Jupyter, OpenCV, scikit-learn, Pandas
- **Programming Languages** : C++, Python, JavaScript, MATLAB, Mathematica, Delphi
- **Tools** : Git, VirtualBox, LTng, Perf, Valgrind
- **CAD/CAE** : SolidWorks, Autodesk Inventor, PTC Creo, CATIA, ANSYS (CFD and Mechanical), Autodesk AutoCAD, Autodesk Moldflow, FeatureCAM, MSC Adams, ZWCAD, Altair HyperMesh, Altair OptiStruct, Materialise Magics, Materialise Mimics, CNC G-code

PROJECTS

Autonomous Robot Navigation

Udacity

C++, Python, ROS, Gazebo, AMCL, gmapping, RTABMap

Apr 2021 – Apr 2022

- Simulated Automated Guided vehicles (AGVs) and Autonomous Mobile Robots (AMRs) in a warehouse environment.
- Mapped a virtual environment, by manually moving an AGV and creating an occupancy grid map. The AGV only had a 2D lidar sensor on it. Localized an AMR using the same map while controlling it manually or autonomously using path planning. Simulated a complete pick-and-drop operation. ROS packages like gmapping, AMCL, and move_base were used.
- Used SLAM to manually move an AMR through a virtual environment and create a loop-closed 3D graph map using RTAB-Map. The AMR used a 3D camera and a 2D lidar sensor to create the map for localization.
- Programmed an AMR to autonomously follow a moving reflective ball target using a 2D camera.
- Used A-star path planning to move a payload through a 2D maze using a KUKA industrial robotic arm as part of [KUKA Robotics Challenge](#).

Self Driving Car subsystem design and integration

Udacity

Python, Jupyter, OpenCV, TensorFlow, Keras, C++, ROS

May 2019 – Aug 2019

- Created a robust image processing pipeline to detect highway lanes in dashcam live-feed. The car's position within lane and lane curvature was calculated using bird's eye view (BEV) and polynomial fitting. LeNet-inspired convolution neural network was also developed to detect and classify 40+ kinds of traffic signs.
- Used a 2D particle filter with GPS, sensor data, and landmark map for sparse localization. Also localized using an extended Kalman filter which fused LIDAR and RADAR sensor data.
- Autonomously drove on a virtual highway with other cars using a Finite State Machine based trajectory planner.
- Implemented a PID controller to maneuver a vehicle around a virtual track using steering, throttle, and brake.
- Used Robot Operating System (ROS) framework to robustly integrate Perception, Localization, Planning, and Control.

Master's projects

Aug 2015 – May 2017

Machine Learning approach to Path Synthesis of crank-rocker mechanisms

- Path tracing mechanisms with optimum transmission angle are synthesized using Artificial Neural Network.
- Input features are calculated using Wavelet transformation to capture the temporal path information.

Motion Synthesis of Planar Four-bar Mechanisms

- Solving Five pose Burmester Motion Synthesis Problem using Kinematic Mapping in an interactive OpenGL based environment implementation in C++.

Motion Planning for a Robot with Two Anthropomorphic 6-DOF Arms

- Inverse Kinematics and Dual Quaternion interpolation based optimal trajectory planning to pick and place objects considering individual arm's workspace and dexterity.

Fracture test analysis for compact tension specimen

- Finite element analysis of a fracture specimen to predict and validate deformations at the crack tip.

Quality Improvement of Aircraft Wing Assembly

- Identifying and correcting the root cause for high rejection rate of final assembly using Pareto Charts, Cause and Effect Diagrams, Control Charts, and Histograms.

Conceptual Design

- Conceptual design of an Ergonomic Nutcracker and Stone Crusher.
- Formulation of product design specification and criteria and generation of the best possible product concept.

Bachelor's projects

Aug 2009 – May 2013

Tessellation Algorithms

- Reviewed algorithms used for the tessellation of Voronoi Diagrams and Delaunay Triangulation and implemented tessellation on a sample set.
- Coding and simulation were done on MATLAB.

Mold Design and Manufacturing

- Designed and manufactured molds for irrigation industry products for Injection Molding Process.
- Flow Analysis results were used to optimize the design and a Pricing Strategy was developed for industry partners.

Deformation Analysis and Validation of Cycle Frame

- Modeled frame elements and calculated the deformations for the model on MATLAB.
- Analyzed the model on ANSYS to validate the results.

Automatic Bartender

- Developed a concept of a sensor-based automatic multiple liquid mixing mechanism and simulated it using PLC (Programmable Logic Control).
- Used Solidworks, PLC ladder logic and Labview.

Remote Controlled Hovercraft

- Designed, analyzed and fabricated a Remote Controlled Hovercraft.
- Modeled the design, performed the lift and thrust calculations and did CFD analysis.
- Used ANSYS CFX, Solidworks, PTC Creo, and Labview.

Computer Aided Manufacturing Projects

- Solved a simplified 2-D Stock Cutting Problem and implemented it through GUI using MATLAB.
- Modeled, Assembled and Simulated the Treadle Sewing Machine using PTC Creo.

Computer Aided Design Projects

- Developed interactive GUI for 3D Transformation of Geometrical Shapes on MATLAB.
- Implemented Hermite Curve, Bezier Curve and B-Splines in 3D on MATLAB.
- Implemented Bilinear Surfaces, Ruled Surfaces, Coons Bicubic Surface Patch, Bezier Surfaces and B-Spline Surfaces in 3D on MATLAB.

Computer Aided Optimization Projects

- Topography optimization of a pressure tank to minimize wall displacement.
- Topology optimization of an automotive control arm to minimize deformation and volume of the component.
- Topology optimization of a clutch pedal to minimize the weight of the part.
- Free size optimization of an aircraft wing rib to minimize weight and maximize stiffness.
- Used Altair Hyperworks, HyperMesh, and OptiStruct.

Human Powered Energy Device

- Designed and Fabricated a Product Concept that harnessed human power.
- Two attachments were designed: Portable Fan and Electric Generator.
- Modeled and Assembled using PTC Creo.

Landing Gear Mechanism

- Analyzed and Fabricated Airplane Landing Gear Mechanism. Analysis was done using MSC Adams.